

3(a)	solid straight line drawn between centre of sphere at X and at Y	B1
3(b)	$p = mv$ or $0.72 = mv$	C1
	$E = \frac{1}{2}mv^2$ or $0.86 = \frac{1}{2}mv^2$	C1
	$(m =) 0.72^2 / (2 \times 0.86) = 0.30 \text{ (kg)}$ or $v = 2E_K / p$ $v = (0.86 \times 2) / 0.72 = 2.4 \text{ (to 2 s.f.)}$ $m = 0.72 / 2.4 = 0.30 \text{ (kg)}$	A1
3(c)	$(\Delta)E = mg(\Delta)h$	C1
	$h = 0.86 / (0.30 \times 9.81)$ $= 0.29 \text{ m}$	A1
3(d)	$\cos \theta = (0.93 - 0.29) / 0.93$ so $\theta = 47^\circ$	A1
3(e)	moment = $(0.30 \times 9.81) \times 0.93 \times (\sin 47^\circ \text{ or } \cos 43^\circ)$ or moment = $(0.30 \times 9.81) \times [0.93^2 - (0.93 - 0.29)^2]^{0.5}$	C1
	$= 2.0 \text{ N m}$	A1

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Question	Answer	Mark
3(f)	there is a <u>resultant</u> force (acting on sphere) or there is a <u>resultant</u> moment (about P acting on pendulum) (so) not in equilibrium	B1

Question	Answer	Marks
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